6 Engineering Desktop Client

DONE | TOTAL: orig ~4d · total ~0d | completed 13 Oct 2026

GOAL

Establish SI-02 as the reusable Engineering Desktop Client, with a reproducible desktop runtime/package baseline and public-interface-only operation against SI-01.

RESULT

- SI-02 is a reusable Engineering Desktop Client software item.
- Normal operator interaction remains IF-04 Web.
- Client/application behaviour remains below JavaFX/docking presentation.
- Java 21 / BentoFX / Windows app-image operation is CI-qualified.
- Lua scripting reuses the same public client/service boundary instead of creating a
- SI-02 design and user guidance are separated into SSD/SDD/SUM roles.

END DEMO

- Start packaged SI-02 and connect it to a running SI-01.
- Observe SYNCING then LIVE, TimingNode state, interpreted registrations, technical
- Exercise registration/simulation and one Lua engineering script.
- Revoke through the GUI, restart SI-01, reconnect and prove DELETED/history recovery.
- Show Device Log, Client Log and Remote Shell as independent engineering surfaces.

DOCUMENTATION

30-UC R-85 | 41-02-SSD R-70 | 43-02-SDD-01 W-80 | 44-02-GPD W-90 | 50-SDE-03 R-90 | 70-02-SUM W-70 | 32-03-IDD R-85 | 60-SVP R-80

DOCUMENTATION / DECISIONS

D01 DONE

Establish SI-02 Engineering Client requirement slice

Establish SI02-REQ-001..008...

D03 after A01 DONE

Complete SI-02 scripting design and user-documentation split

Complete the current SI-02...

APPLICATION / PRODUCT

D02 after D01 DONE

Select Engineering Client desktop stack, client boundary and packaging

Select Java 21, JavaFX 21,...

A01 after D01 D02 DONE

Restructure the Engineering Client around the selected workbench/services

Completed through Java issue...

A02 after A01 DONE

Add client-side Lua engineering automation and capability diagnostics

Add the local Lua engineering...

VERIFICATION / TEST

V02 after A02 D03 DONE

Verify Lua/capability behaviour with automated client tests and CI

Automated SI-02 tests and...

V01 after A01 A02 V02 DONE

Verify packaged Engineering Client connection, sync/reconnect and representative workbench behaviour

Accepted through repeated...

PLANNING CHANGES

- 2026-10-13 — Close Step 6 on repeated real SI-02/SI-01 exploratory running-system use rather than require a second artificial line-by-line execution of VC-ST1-003. Record V01 as accepted integrated-client evidence without claiming a formally completed checklist; keep VC-ST1-003 as reusable regression guidance for later verification.
- 2026-10-13 — Record the post-A01 SI-02 extension as part of the Step-6 baseline: source-driven capability diagnostics, client-side Lua engineering scripting, focused Lua SDD and Engineering Client SUM are implemented and automatically verified. V01 / VC-ST1-003 remains the only formal closeout gate; reduce the remaining estimate to 0.5 project day and do not report Step 6 complete before the real packaged-GUI run.
- 2026-10-09 — Complete A01 through Java PR #402: SI-02 now runs on Java 21 / JavaFX 21 with BentoFX, an instance-scoped system context, real dockable panes and a CI-verified Windows jpackage app-image. V01 is next; with only that running-system verification remaining, the Step-6 remainder is re-estimated at 1 project day.

PLANNING CHANGES (CONT.)

- 2026-10-09 — Classify the reusable Engineering Desktop Client as SI-02. Keep UC-009 as its system use case, IF-03 as its primary SI-01 boundary and IF-04 Web as the normal operator path; UC-008 remains outside the current baseline.
- 2026-10-09 — Close Step 5 on SI-01/server evidence and assign the real GUI revoke/DELETED/reconnect/recovery scenario to Step 6 V01, releasing A01.
- 2026-10-09 — Select Java 21, JavaFX 21, BentoFX 0.16.0, JDK HTTP/WebSocket, Jackson, Maven and a jpackage app-image as the Step-6 desktop baseline.
- 2026-10-09 — Reframe Step 6 from an operator desktop GUI to an Engineering Desktop Client; normal operator use remains IF-04 Web.

PLANNING CHANGES (CONT.)

- 2026-10-07 — Promote the first SI-02 Draft requirement slice before selecting GUI technology; implementation remains gated on the then-current Step-5 V06 plan.